Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/10/2005
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Project Title:	Control of Bacterial Biofilm Formation in Space Exploration		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	25755-9320	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Bacteria in nature and disease often choose a mode of growth known as a biofilm, which consists of numerous aggregating microorganisms (microcolonies) attached to a surface. Biofilm formation is a leading cause of bacterial pneumonia which can be acquired by astronauts during a long-term space mission and by patients with compromised immune systems such as those with cystic fibrosis. A capsular material called mucoid exopolysaccharide, or alginate is important in the formation of biofilms because it provides a key structural component for in vivo biofilms. The objective of this proposal is to study the genetic regulation of alginate in biofilm formation. The model organism is Pseudomonas aeruginosa, a ubiquitous biofilm-forming bacterium capable of displaying two distinctive phenotypes: a nonmucoid or mucoid colony morphology. The mucoid form, which is due to the overproduction of alginate, is a clinical indicator for a poor prognosis. While the two morphologies can be made inter-changeable under the laboratory conditions, little is known about the molecular mechanisms governing this process. Using genetics and functional genomics approaches, we

Task Description:	will search for the biofilm “switches” on the genomes. The identified genes may become targets for the development of anti-biofilm therapeutics. The specific aims of this project are i) to identify and characterize the genes in the alginate regulation, ii) to elucidate the underlying mechanisms, and iii) determine what/how environmental signals modulate the expression of these regulators. Our long-term goal is to target bacterial own genes for the control of the space biofilm formation. The objective of the proposed research is to identify and characterize the biofilm-controlling genetic elements from the genomes of P. aeruginosa. It is our expectation that the approaches used here will lead us to identify the genes that control the formation of the mucoid biofilms. We will also provide new information regarding how these genes control the phenotypic switch, and what/how environmental signals may stimulate or inhibit the expression of these genes. These results will be significant, because they will provide genetic targets for anti-biofilm therapeutics, and aid in the future development of the gene-based space medicine. In addition, it is expected that the results will advance the emerging field of microbial biofilms.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The major Earth benefits of this proposal are to provide the basis for the future development of the new genetically defined drug with improved specificity and function and very little side effect.
Task Progress:	1. Development of a transposon mutagenesis approach to identifying the alginate regulators. Since the transposon insertion mutations are null alleles, we must examine whether there will be polar effect on the immediate downstream genes by complementation and gene inactivation. We also want to prove that the mucoid mutant phenotype is due to the transposon insertion by a genetic backcross. The mutants generated will be tested for which pathway the mutation affects by constructing the transcriptional fusion using the alginate-specific genes such as algU, rpoN, algR, and algB fused with reporter systems such as gfp or lacZ. Using forward genetic analysis approaches, four candidate genes have been identified in the PAO1 genome and named mucoidy inhibitor genes A-D (muiA-D). To further dissect the role of these candidates, we will genetically and biochemically characterize MuiA-D and study how their expression in mucoid cells causes this phenotypic conversion. Each of these genes will be inactivated from bacterial genome to test the effect on the phenotype. To further characterize these mutants, we will test the specific and global effects of the mutation using the real-time PCR, microarray and 2-D gel electrophoresis. 2. Characterization of suppressor mutation that causes the phenotypic reversion from a mucoid to nonmucoid morphology. We will determine the reversion rates of these defined mutants using the Luria-Delbruck fluctuation test. We will continue to search for the original and suppressor mutations in PAO579 and PAO579NM. Once these mutations are found, we will change them from mutant form back to wild type or vice versa to confirm with the phenotype that they are indeed the causative mutations. The identification of these mutations will help to understand how they affect the phenotypic switch from nonmucoid ? mucoid ? nonmucoid. 3. Development of an efficient recombination system for gene inactivation in P. aeruginosa. In addition to the functional genomics analysis as outlined above, developing a P. aeruginosa strain with an increased frequency of homologous recombination using the linear PCR molecules as templates will be useful. One application is the use of this recombinant to investigate the effect of radiation on recombination and DNA repair systems in P. aeruginosa. While current knowledge on the DNA repair system is derived mainly from the studies on E. coli, it would be interesting to see whether there is difference and similarity in terms of the DNA repair and recombination systems when P. aeruginosa is irradiated. It is clear that space irradiation can chemically damage DNA due to its absorption of the energy of the photons. DNA damage can cause mutations, many of which may be deleterious or lethal. Therefore, development of this recombinant strain may allow us to expand our research to a new field of space biology.
Bibliography Type:	Description: (Last Updated: 04/09/2020)
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